EX NO:3.2 CREATE 3D MODEL FROMvORTHOGRAPHIC PROJECTION

AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

Step 1: Sketching Base Setup

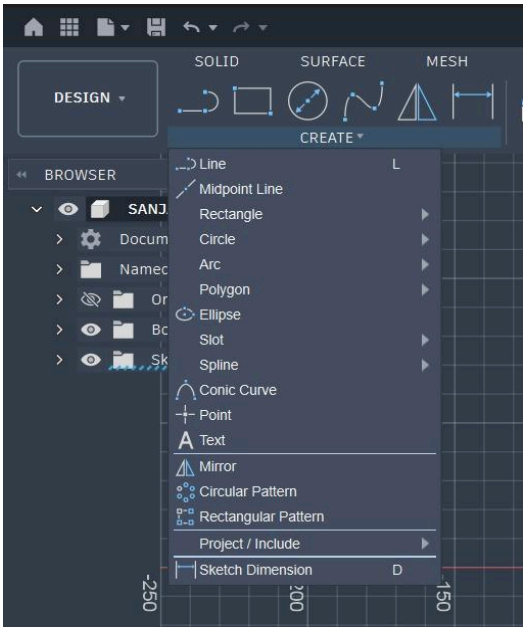
1. Open Autodesk Fusion 360 and create a **New Design**.
2. Click **Create Sketch** and select the **XY Plane** (Top Plane).

Step 2: Base Profile Creation (Dimensions & Constraints)

1. **Draw Base Rectangle:**
    - Select Rectangle tool (Center Rectangle or 2-Point Rectangle).
    - **Dimensions:** Set Length = **80 mm**, Width = **50 mm**.
    - **Constraints:** Apply **Coincident Constraint** to align the center/corner of the rectangle with the Origin point $(0, 0)$. Apply **Horizontal/Vertical Constraints** to ensure all sides remain parallel to axes.
2. **Add Circles / Slots (if required by drawing):**
    - Select Circle tool (C).
    - **Dimensions:** Draw circles of Diameter $\varnothing 12\text{ mm}$ (or as per input sheet).
    - **Constraints:** Apply **Dimension Constraint** (D) to set center-to-center distance (e.g., $40mm$ apart) and apply **Equal Constraint** if multiple circles share the same diameter.
3. **Finish Sketch:** Click **Finish Sketch** in the top ribbon once the fully constrained sketch turns black.

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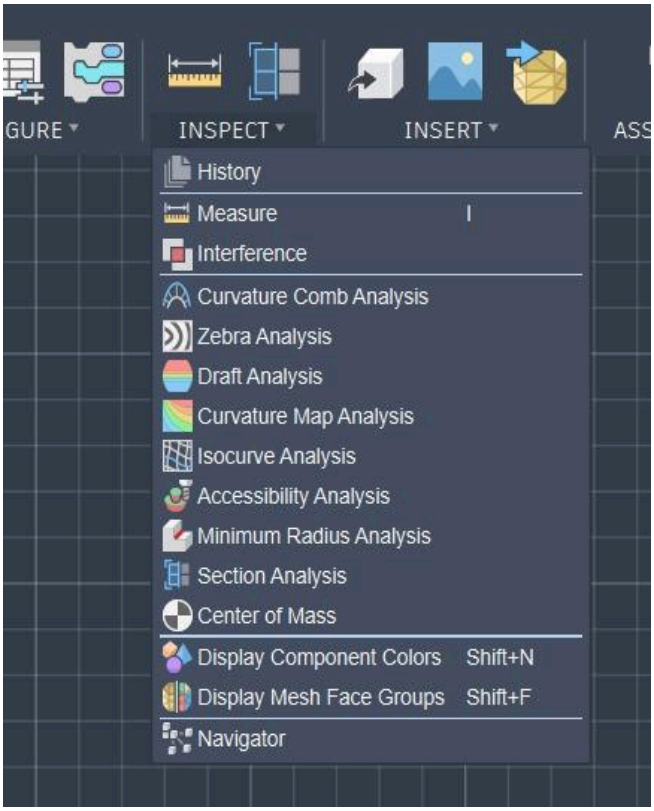
3.Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch.



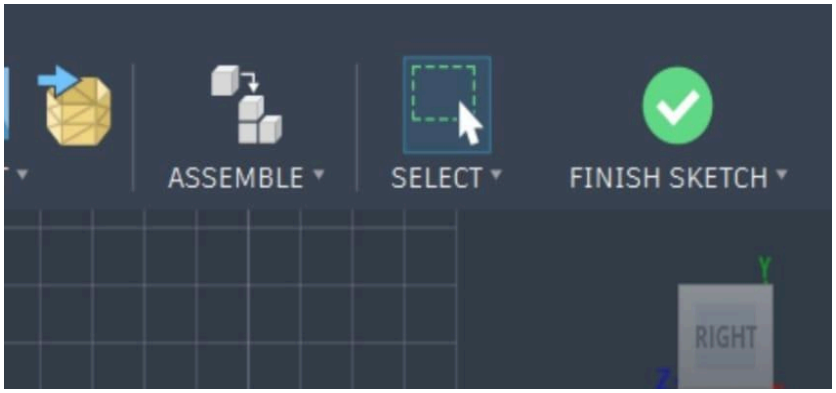
4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch.



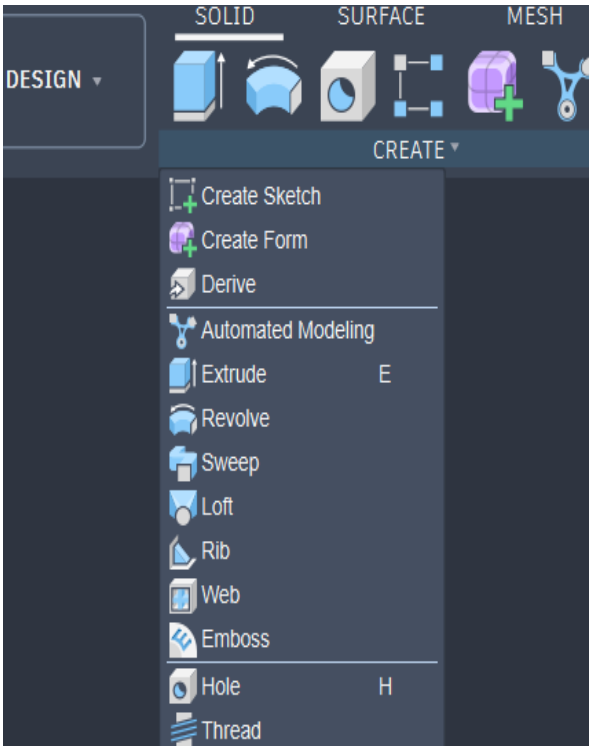
5. Use inspect feature to measure the dimensions of sketch.



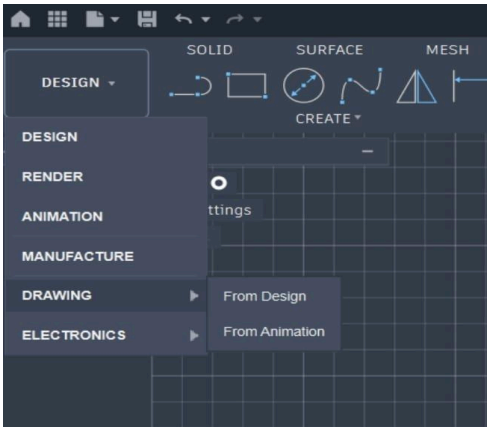
6. After finishing the sketch use finish sketch option to finalize the sketch.



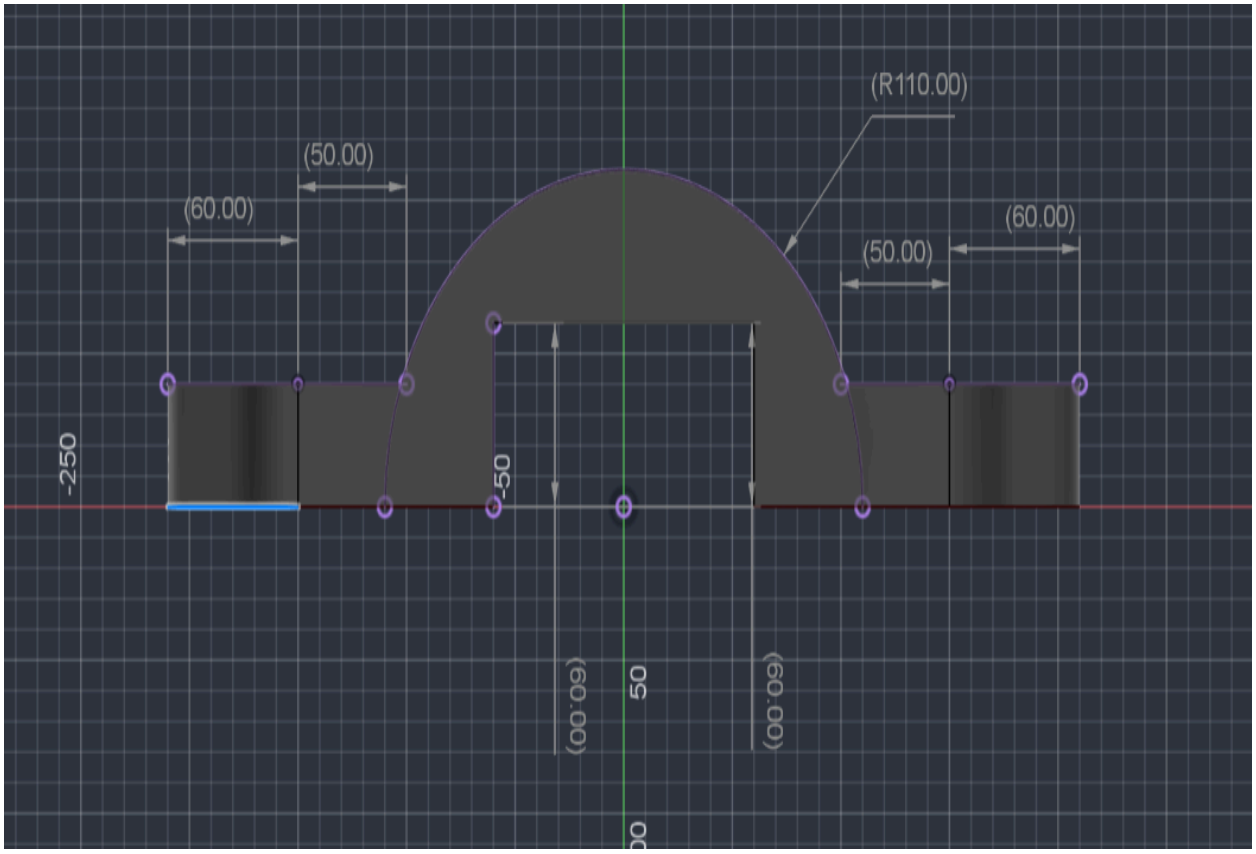
7. Use extrude option to convert 3D model from sketch
8. Select the base profile.
9. Set **Distance** = **15 mm** (Base Thickness) and Operation = **New Body**.



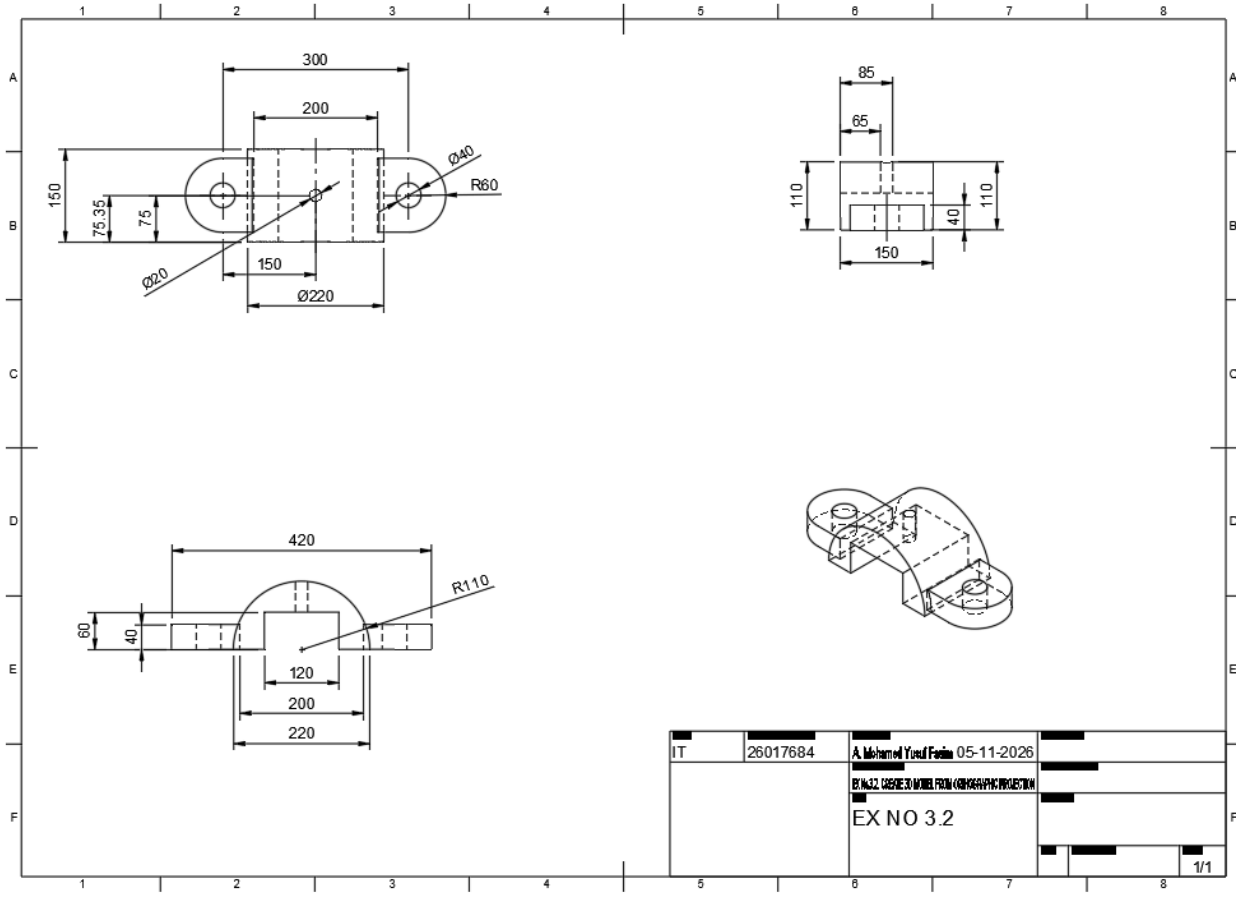
8. After finishing the model convert to drawing sheet per ISO standard.



INPUT:



OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projectio